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MAHATMA GANDHI UNIVERSITY, KOTTAYAM
MGU-UGP (HONOURS) Regular EXAMINATION October 2025
Third SEMESTER
Discipline Specific Elective (DSE) - MG3DSECHE201 – INTRODUCTION TO NANOSCIENCE
(2024 ADMISSION ONWARDS)

Instruction to the Examiners:

- This scheme is only a guideline for valuation. Any other attempt which is equally valid and correct may be given due marks.
- Exceptions are there in some cases and if the answer is complete full marks may be given even if the required number of valuation points could not be identified.
- Some of the answers given in the scheme are elaborate and this is to help the examiner in their valuation. Please do not expect all those points in the answers of the students.

Part A
Short Answer Type Questions
Answer any 5 questions, Each question carries 4 marks

1. How do geckos' feet exhibit nanoscale features that enable them to stick to surfaces?

Gecko feet have millions of microscopic hairs called setae, each branching into nanoscale spatulae that increase surface contact. These tiny structures generate strong van der Waals forces between the gecko's foot and the surface, allowing it to stick without glue.

(Relevant explanation - 4 marks)

2. List four applications of TEM.

a) TEM provides detailed images of atomic arrangements and crystal structures in materials.
b) It is used to study nanomaterials such as nanoparticles and nanowires to understand their size and internal structure.
c) In biology and medicine, TEM reveals detailed views of cells, viruses, and bacteria to study diseases and cell functions.
d) It helps analyze and improve metals, alloys, and other materials for advanced industrial and electronic applications.

(Any four applications of TEM – 4 marks)

3. Explain the concept of quantum dots with an example.

Quantum dots are tiny semiconductor particles, only a few nanometres in size, that have unique optical and electronic properties due to quantum confinement of electrons. Their colour of light emission depends on their size. Examples of quantum dots include cadmium sulphide (CdS), cadmium selenide (CdSe), cadmium telluride (CdTe), and indium phosphide (InP) nanoparticles.

(Definition and explanation of quantum dots – 3 marks, any one example – 1 mark)

4. How does nanotechnology contribute to solar energy harvesting?

Nanotechnology improves solar cell harvesting by increasing light absorption, charge transport, and energy conversion efficiency. Nanomaterials like quantum dots, carbon nanotubes, and metal nanoparticles capture a broader range of sunlight and reduce energy loss. They also make solar cells thinner, flexible, and more cost-effective, enabling applications in wearable and portable energy devices.

(Any valid 3 points - 4 marks)

5. Explain the characteristics and potential applications of one-dimensional nanoparticle

One-dimensional nanoparticles are nanomaterials that have one dimension (diameter) in the nanometre range while the other two dimensions are much larger, giving them a wire-, rod-, or tube-like shape. Their high aspect ratio gives them excellent electrical conductivity, mechanical strength, and directional charge or heat transport properties. They can be used in nanoelectronics, sensors, energy storage devices, and catalysts.

(two characteristics – 3 marks, one application – 1 mark)

6. Recall what carbon dots are. Mention one method of their synthesis and list the raw materials used as carbon sources in their synthesis.

Carbon dots are quasi spherical nanoparticles primarily made of carbon, however they may also contain hydrogen, oxygen, nitrogen or sulphur

Hydrothermal method for synthesis of carbon dots: In the hydrothermal method, a carbon-rich precursor such as citric acid or glucose is dissolved in water and sealed in a Teflon-lined autoclave. The mixture is heated at around 150–200 °C for several hours, allowing carbonization and formation of carbon dots. After cooling, the resulting solution is filtered or centrifuged to obtain purified carbon dots.

Carbon source: citric acid, glucose, orange peel or any other organic-rich biomass.

(Definition of carbon dots – 1 mark, any one method for synthesis of carbon dot – 2 marks, any one carbon source – 1 mark)

7. How do nanomaterials facilitate data storage?

Nanomaterials facilitate data storage by enabling the creation of smaller, faster, and more efficient storage devices. Magnetic nanoparticles can store data at the nanoscale by aligning their magnetic moments, greatly increasing storage density. Nanostructured materials like carbon nanotubes and graphene also enhance the speed and durability of memory devices by improving electrical conductivity and reducing energy loss.

(Any valid 3 points - 4 marks)

Part B
Short Essay Type Questions
Answer any 5 questions, Each question carries 7 marks

8. Describe how magnetic nanoparticles help in environmental remediation.

Magnetic nanoparticles play a key role in environmental remediation due to their large surface area, reactivity, and easy magnetic separation. They can adsorb or degrade pollutants such as heavy metals, dyes, and organic contaminants from water and soil. Functionalized magnetic nanoparticles can be tailored with specific coatings to selectively capture target pollutants. After treatment, they can be easily removed from the environment using an external magnetic field, minimizing secondary pollution. Iron oxide nanoparticles, for example, are widely used to remove arsenic, lead, and chromium from contaminated water. Their reusability and high efficiency make them a sustainable and cost-effective tool for cleaning up environmental pollutants.

(Any valid 5 points - 7 marks)

9. Discuss the role of nanotechnology in developing biosensors. Illustrate with examples.

A nano-biosensor is a biosensor at a nanoscale. It has a biological component that acts as a sensor and an electronic component that detects and transmits signal. By operating at the nanoscale, these sensors can have enhanced sensitivity and selectivity, allowing for the detection of tiny amounts of analytes at high speed, such as for diagnosing diseases or monitoring food safety. Examples: Carbon nanotube-based sensors for detecting bacteria and chemicals, DNA-based fluorescent sensors for detecting biological molecules.

(Relevant explanation on Biosensors (Any valid 5 points with examples)- 7 marks; Explanation only - 5 marks)

10. Discuss the advantages of nanotechnology-based diagnosis over conventional methods.

Nanotechnology-based diagnostic methods offer several advantages over conventional approaches. They provide high sensitivity and specificity, allowing detection of minute quantities of biomolecules and enabling early disease diagnosis. The tests are rapid and efficient, significantly reducing the time required for obtaining results. Owing to miniaturization, nanodiagnostic devices are portable and suitable for point-of-care testing. They also allow simultaneous detection of multiple biomarkers, giving a more comprehensive understanding of disease states. Many of these techniques are non-invasive, using samples such as saliva or urine instead of tissue biopsies. Nanoparticles enhance signal generation, leading to greater accuracy and reliability. These systems can also be integrated with digital and smartphone-based platforms for real-time monitoring. Furthermore, they require smaller sample volumes, making them cost-effective, and support personalized medicine by enabling precise biomarker identification for individualized treatment strategies.

(Any valid 5 points - 7 marks)

11. Write the working principle and procedure of chemical precipitation method for nanomaterial synthesis. How pH, temperature and precursor concentration influence particle size and morphology of synthesised nanomaterials?

The chemical precipitation method for nanoparticle synthesis involves adding a precipitating agent to a metal salt solution under controlled conditions, causing nanoparticles to form as insoluble precipitates. pH, temperature, and precursor concentration influence nanoparticle size and morphology by controlling nucleation, growth rates, and particle stability.

(Principle – 2 marks, Procedure – 3 marks, influence of pH, temp and precursor concentration – 2 marks)

12. Discuss the principle of AFM. Evaluate its importance in the characterization of nanomaterials.

The principle of Atomic Force Microscopy (AFM) is based on scanning a sharp tip on a flexible cantilever over a surface and detecting the intermolecular forces between the tip and the sample. As the tip scans, it experiences forces (attractive or repulsive) that cause the cantilever to bend or change its oscillation. This deflection is detected by a laser beam that reflects off the back of the cantilever onto a position-sensitive photodiode, which translates the movement into a 3D image of the sample's surface topography. AFM can image surfaces at the nanometer and even atomic scale, offering exceptional detail. Unlike many microscopes, AFM

provides true 3D topographical maps of the sample surface. AFM can operate in air, vacuum, or liquid, making it suitable for both biological and material samples.

(Principle of AFM- 4 marks; Importance of AFM - any 3 points - 3 marks)

13. Discuss Richard Feynman's hypothesis on atomic-scale manipulation and explain its historical significance, and how it laid the foundation for the development of nanotechnology.

The talk was given in 1959 at Caltech by physicist Richard P. Feynman. The title of the talk was "There's Plenty of Room at the Bottom." Feynman suggested that scientists could build machines and materials atom by atom. He imagined tiny machines that could make even smaller machines, like today's nanorobots. He talked about storing large amounts of information in very small space. His ideas inspired modern fields such as nanoscience, nanomedicine, and microelectronics. Feynman's vision came true decades later with tools like the scanning tunnelling microscope and nanofabrication techniques.

(**Any valid 5 points - 7 marks)**

14. Explain the importance of UV Visible spectra and IR spectra in the characterization of nanomaterials

UV-Vis spectroscopy helps find the size and concentration of nanomaterials by measuring light absorption. Nanoparticles like gold and silver show a special feature called surface plasmon resonance (SPR), where free electrons resonate with electromagnetic radiation, causing strong absorption peaks. SPR peaks in UV-Visible spectra reveal the size, shape, and stability of nanoparticles in solution.

IR spectroscopy identifies the chemical bonds and groups on nanomaterial surfaces. It helps confirm if the nanomaterial contains specific molecules or coating substances. IR gives information about how atoms in nanomaterials are bonded and changes during reactions

(Relevant explanation, UV – 3.5 marks, IR – 3.5 marks)

Part C
Essay Type Questions, Answer any 1 question, Each question carries 15 marks

15. Compare SEM and TEM

1. SEM shows detailed 3D images of the surface of a sample, while TEM shows 2D images of the internal structure.
2. SEM works well with thicker samples and requires simpler preparation, but TEM needs very thin samples and more complex preparation.
3. SEM has lower resolution (around 1-10 nm), whereas TEM can achieve much higher resolution, even below 1 nanometer.
4. SEM provides faster imaging and is easier to operate; TEM takes longer and requires specialized training.
5. SEM is generally less expensive and covers larger sample areas, while TEM is costlier but gives atomic-scale details.

(Any 5 points of comparison – 15 marks)

16. Explain the top-down approaches used in nanochemistry. Discuss its methods, principle, advantages and limitations with suitable examples.

Top-down approaches in nanochemistry involve breaking down bulk materials into nanoscale particles.
Methods: Mechanical milling, Laser ablation , Arc discharge etc.
Advantages:
1.Can produce large quantities of nanomaterials. 2. Well-established techniques easily scaled up for industrial use. 3. Suitable for a variety of materials including metals and ceramics.
Limitations:
1.Difficult to control particle size and shape precisely. 2. Introduction of defects and impurities during mechanical processing. 3. Requires expensive equipment and high energy input for some methods.
Examples: silver and gold nanoparticles made by laser ablation and carbon nanotubes produced by arc discharge methods.

(Principle- 2 marks; Discussion of any two top-down methods-3 marks for each method; Advantages-3 marks; Limitations-3 marks Examples-1 mark)